HW_Assignment_1_Fall2026

September 28, 2026 4:31 PM



HW_Assign
ment_1_...

ENEL 573 Computer Networks: HW Assignment 1 (Fall 2026)

(Due on Sept 28th, at 11:59pm MST)

Name: Temuulen Ankhbayar **UCID:** 30188632 **Score:** _____

Problem 1 (20 points). Consider two hosts A and B separated by 2 nodes (switches or routers). Host A wants to send a file of size $M = 15$ Mbytes over to host B. Each link has the same data rate $C = 12$ Mbit/s.

a) Assume **"message" switching**, how long would it take for the whole file to be received by host B? Write the formula giving the time in terms of C, M, and possibly other parameters and explain your assumptions (note: M and C are in different units). **(10 points)**

b) Assume **packet switching** and that all packets have the same size $L = 9600$ bits, how long would it take for the whole file to be received by host B? Explain your assumptions. Write the formula giving the time in terms of C, M, and possibly other parameters. **(10 points)**

a) $h = 3$

$M = 15\,MB = 120\,Mb$

$C = 12\,Mb/s$

$T = h\frac{M}{C} = (3)\frac{120\,Mb}{12\,Mb/s} = 30s$

Assume: Store and Forward
d_{prop} is negligible
d_{proc} is negligible

b) $L = 9600\,b$

$P = \frac{M}{L} = \frac{120\,Mb}{9600\,b} = 12\,500$

$T = (P + h - 1)\left(\frac{L}{C}\right) = (12500 + 3 - 1)\left(\frac{9600\,b}{12\,Mb/s}\right) = 10s$

Assume: Store and Forward for packets
d_{prop} is negligible
d_{proc} is negligible

Problem 2 (30 points) In the following 6 questions, we are sending a 30 Mbit MP3 file from a source host to a destination host. All links in the path between source and destination have a transmission rate of 10 Mbps. Assume that the propagation speed is $2 * 10^8$ meters/sec, and the distance between source and destination is 10,000 km.

Note: For each multiple-choice question below, please provide the detailed derivation process of how your answer is obtained. Only providing correct answer selection without justification will gain half of the full marks for each question.

1) Initially suppose there is only one link between source and destination. Suppose that **message switching (circuit switching)** is used, with the message consisting of the entire MP3 file. The transmission delay is **(5 points)**

a) 50 milliseconds
b) 3 seconds
c) 3.05 seconds
d) none of the above.

$F = 30\,Mb$
$R = 10\,Mb/s$
$s = 2 \times 10^8\,m/s$
$D = 10\,000\,km = 10^7\,m$

$d_{trans} = \frac{F}{R} = \frac{30\,Mb}{10\,Mb/s} = 3s$

2) Referring to the above question, the end-to-end delay (transmission delay + propagation delay) from the time instant that host A starts to send the file to the instant that host B receives the entire file is **(5 points)**

a) 6 seconds
b) 3.05 seconds
c) 3 seconds
d) none of the above

$d_{prop} = \frac{D}{s} = \frac{10^7 m}{2 \times 10^8 m/s} = 0.05s$

$d_{end\ to\ end} = d_{tran} + d_{prop} = 3s + 0.05s = 3.05s$

3) Referring to the above question, how many bits will the source have transmitted when the first bit arrives at the destination? **(5 points)**

a) 500,000 bits
b) 1 bit
c) 30,000,000 bits
d) none of the above

$R\, d_{prop} = (10\ Mb/s)(0.05s) = 0.5\ Mb$

4) Now suppose there are two links between source and destination, with one router connecting the two links. Each link is 5,000 km long. **Again, suppose the MP3 file is sent as one message.** Suppose there is no congestion so that the message is transmitted onto the second link as soon as the router receives the entire message. The end-to-end delay (transmission delay + propagation delay) is **(5 points)**

a) 6.05 seconds
b) 6.1 seconds
c) 3.05 seconds
d) none of the above

$d_{prop} = \frac{D}{s} = \frac{5 \times 10^6 m}{2 \times 10^8 m/s} = 0.025s$

$d_{end\ to\ end} = 2(d_{prop} + d_{tran}) = 2(3s + 0.025s) = 6.05s$

5*) Now suppose that the MP3 file is broken into 3 packets, each of 10 Mbits. Ignore headers that may be added to these packets. Also ignore router processing delays (i.e., the time each router takes to process packet header). Assuming **store-and-forward packet switching** at the router, the end-to-end delay for transmitting the entire file is **(5 points)**

a) 3.05 seconds
b) 4.05 seconds
c) 6.05 seconds
d) none of the above

$d_{trans} = \frac{10\,Mb}{10\,Mb/s} = 1s$

$T = (P + h - 1)d_{trans} + h d_{prop} = (3 + 2 - 1)(1s) + (2)(0.025s) = 4.05s$

6) Now suppose there is only one link between source and destination, and there are 10 frequency division multiplexing (FDM) channels in the link. The MP3 file is sent over one of the channels with circuit switching. The end-to-end delay is **(5 points)**

a) 30.05 seconds
b) 30 seconds
c) 300 microseconds
d) none of the above

$\frac{R}{10} = \frac{10\,Mb/s}{10} = 1\,Mb/s$

$d_{prop} = \frac{10^7\,m}{2 \times 10^8\,m/s} = 0.05s$

$d_{trans} = \frac{30\,Mb}{1\,Mb/s} = 30s$

$d_{end\ to\ end} = d_{prop} + d_{trans} = 0.05s + 30s = 30.05s$

Problem 3 (20 points). Consider two hosts, Hosts A and B, connected by a single link of rate R bps. Suppose that the two hosts are separated by m meters, and suppose the propagation speed along the link is s meters/sec. Host A is to send a packet of size L bits to Host B.

a. Express the propagation delay, d_{prop} in terms of m and s. **(2 points)**

b. Determine the transmission time of the packet, d_{trans} in terms of L and R. **(3 points)**

c. Ignoring processing and queuing delays, obtain an expression for the end-to-end delay. **(3 points)**

d. Suppose Host A begins to transmit the packet at time $t = 0$. At time $t = d_{trans}$, where is the last bit of the packet? **(3 points)**

e. Suppose d_{prop} is greater than d_{trans}. At time $t = d_{trans}$, where is the first bit of the packet? **(3 points)**

f. Suppose d_{prop} is less than d_{trans}. At time $t = d_{trans}$, where is the first bit of the packet? **(3 points)**

g. Suppose $s = 2.5*10^8$ meters, $L = 1000$ bits, and $R = 284$ kbps. Find the distance m so that d_{prop} equals d_{trans}. **(3 points)**

a) $d_{prop} = \frac{m}{s}$

b) $d_{tran} = \frac{L}{R}$

c) $d_{end\ to\ end} = d_{prop} + d_{tran} = \frac{m}{s} + \frac{L}{R}$

d) in the link

e) in the link

f) at Host B

g) $d_{prop} = d_{tran} \longrightarrow \frac{m}{s} = \frac{L}{R} \longrightarrow m = \frac{Ls}{R} = \frac{(1000)(2.5 \times 10^8)}{284\ kb/s} = 880\ 282\ m$

Problem 4 (15 points): Consider the discussion in Section 1.3 of statistical multiplexing in which an example is provided with a 1 Mbps link. Users are generating data at a rate of 100 kbps when busy, but are busy generating data only with probability $p = 0.1$. Suppose that the 1 Mbps link is replaced by a 1 Gbps link.

a) What is N, the maximum number of users that can be supported simultaneously under circuit switching? **(8 points)**

b) Now consider packet switching and a user population of M users. Give a formula (in terms of p, M, N) for the probability that more than N users are sending data. **(7 points)**

a) $N = \frac{1\ Gb/s}{100\ kb/s} = 10\ 000\ users$

b) $P(X>N) = \sum_{n=N+1}^{M} \frac{M!}{n!(M-n)!} p^n (1-p)^{M-n}$

Problem 5 (15 points): Consider a packet of length L which begins at end system A, travels over one link to a packet switch, and travels from the packet switch over a second link to a destination end system. Let d_i, s_i and R_i denote the length, propagation speed, and the transmission rate of link i, for i=1 ,2. The packet switch delays each packet by d_{proc}. Assuming no queuing delays, in terms of d_i, s_i and R_i (i=1,2), and L, what is the total end-to-end delay for the packet? **(10 points)** Suppose now the packet is 1,000 bytes, the propagation speed on both links is $2.5*10^8$ *m/s*, the transmission rates of both links is 1 Mbps, the packet length is 1,000 bytes, the packet switch processing delay is 1 *msec*, the length of the first link is 4,000 km, and the length of the last link is 1,000 km. For these values, what is the end-to-end delay? **(5 points)**

$$d_{end\ to\ end} = \underbrace{\frac{L}{R_1}}_{d_{trans,1}} + \underbrace{\frac{d_1}{s_1}}_{d_{prop,1}} + d_{proc} + \underbrace{\frac{L}{R_2}}_{d_{trans,2}} + \underbrace{\frac{d_2}{s_2}}_{d_{prop,2}}$$

$$= \frac{8000\,b}{1\,Mb/s} + \frac{4\times10^6\,m}{2.5\times10^8\,m/s} + 0.001s + \frac{8000\,b}{1\,Mb/s} + \frac{1\times10^6\,m}{2.5\times10^8\,m/s}$$

$$= 37ms$$